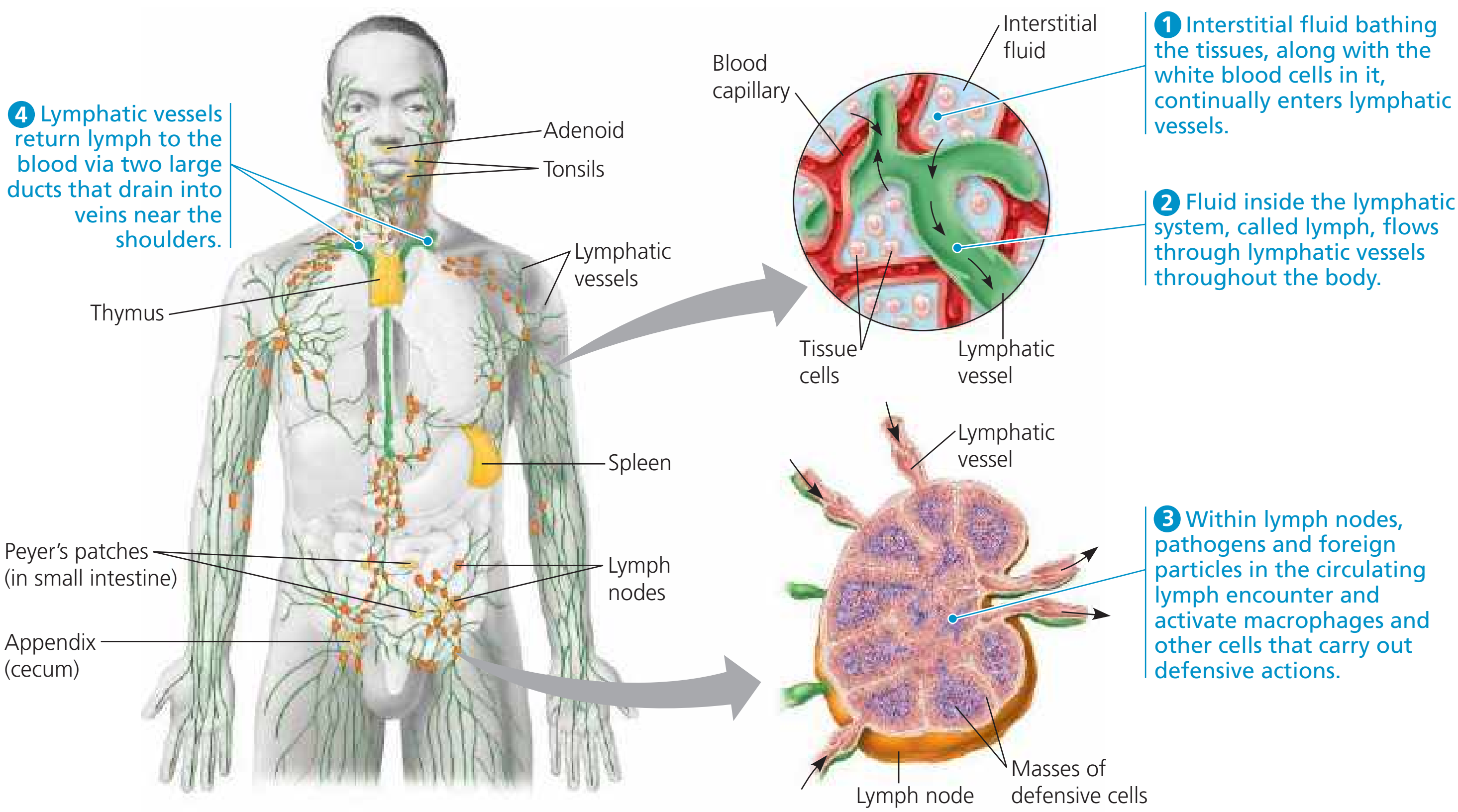
▼ **Figure 43.6 The human lymphatic system.** The lymphatic system consists of lymphatic vessels (shown in green), through which lymph travels, and structures that trap foreign substances. These structures include lymph nodes (orange) and lymphoid organs (yellow): the adenoids, tonsils, thymus, spleen, Peyer's patches, and appendix. Steps 1–4 trace the flow of lymph and illustrate the role of lymph nodes in activating adaptive immunity. (Concept 42.3 describes the relationship between the lymphatic and circulatory systems.)



A local inflammatory response begins when activated macrophages discharge *cytokines*, signaling molecules that recruit neutrophils to the site of injury or infection. In addition, mast cells release the signaling molecule **histamine** at sites of damage. Histamine triggers nearby blood vessels to dilate and become more permeable. The resulting increase in local blood supply produces the redness and increased skin temperature typical of the inflammatory response (from the Latin *inflammare*, to set on fire).

Cycles of signaling and response continue the process of inflammation. Activated complement proteins promote further histamine release, attracting more phagocytic cells to the site of injury and infection. At the same time, enhanced blood flow to the site helps deliver antimicrobial peptides, which, as in insects, typically kill or inactivate pathogens by disrupting membrane integrity. The result is an accumulation of *pus*, a fluid rich in white blood cells, dead pathogens, and debris from damaged tissue.

At the end of the local inflammatory response, pus and excess fluid are taken up as lymph, the fluid transported in the network of vessels known as the lymphatic system **(Figure 43.6)**. Small organs called *lymph nodes* scattered within the lymphatic system contain macrophages, which engulf pathogens that enter the lymph from the interstitial fluid. Dendritic cells reside outside the lymphatic system but migrate to the lymph nodes after interacting with pathogens. Within the lymph nodes, dendritic cells interact with other immune cells, stimulating adaptive immunity.

Systemic and Chronic Inflammation

A minor injury or infection causes a local inflammatory response, but more extensive tissue damage or infection may lead to a response that is systemic (throughout the body). Cells in injured or infected tissue often secrete molecules that stimulate the release of additional neutrophils from the bone marrow. In the case of a severe infection, such as meningitis or appendicitis, the number of white blood cells in the bloodstream may increase severalfold within only a few hours.

A systemic inflammatory response sometimes involves fever. In response to certain pathogens, substances released by activated macrophages cause the body's thermostat to reset to a higher temperature (see Concept 40.3). The benefits of the resulting fever are still a subject of debate. One hypothesis is that an elevated body temperature may enhance phagocytosis and, by speeding up chemical reactions, accelerate tissue repair.

Certain bacterial infections can induce an overwhelming systemic inflammatory response, leading to a life-threatening condition called *septic shock*. Characterized by very high